

Chapter 29

Simplifying Rational Expressions



1) For all $x \neq -3$, which of the following is equivalent to

$$\frac{x^2+4x+3}{x+3} ?$$

- A) $x + 5$ B) $x + 1$ C) $x - 3$ D) $x - 4$ E) $2x$

2) Which of the following could be the solution to the following equation:

$$\frac{1}{3a} + \frac{3a}{2} = \frac{3}{3a} ?$$

- A) $\frac{3}{5}$ B) $\frac{1}{4}$ C) $\frac{1}{8}$ D) $\frac{2}{9}$ E) $\frac{2}{3}$

3) What is the solution set of the equation $\sqrt{x-3} = 5$

- A) 77 B) 28 C) ± 8 D) ± 9 E) 22

4) Which values of x are solutions to the equation $x^2 - 8x + 15 = 0$?

- A) 5 and -7
B) 2 and 6
C) 3 and 5
D) -15 and 1
E) -4 and -8

1	2	3	4	5	6	7	8	9	10
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5) For $x \neq 1$, which one of the following is equivalent to

$$\frac{6x^2 - 15x + 9}{x - 1}?$$

A) $\frac{(3x-2)(2x-5)}{x-1}$

B) $\frac{x-3}{x-1}$

C) $3(2x - 3)$

D) $30x$

E) $2x + 5$

6) Which one of the following values of M makes the following equation true?

$$\frac{4}{M} + \frac{3}{10} = \frac{4}{5}$$

A) 8

B) 10

C) 12

D) 16

E) 20

7) If $4Q^2 + 3 = 12Q - 6$, what is the value of Q ?

A) $\frac{3}{5}$

B) $\frac{4}{5}$

C) $-\frac{3}{5}$

D) $\frac{3}{2}$

E) $-\frac{1}{2}$

8) Simplify the following expression; $\frac{4}{3A} + \frac{3}{4A}$

A) $\frac{4A}{3}$

B) $\frac{25}{12A}$

C) $8A$

D) $\frac{16}{9A}$

E) $7A$

Solutions Chapter 29

Simplifying Rational Expressions

- | | |
|------|------|
| 1) B | 5) C |
| 2) E | 6) A |
| 3) B | 7) D |
| 4) C | 8) B |

1) To simplify, factor the numerator. If there is a trinomial in the form $ax^2 + bx + c$ on the math ACT, factor, and it will almost always lead to the correct solution. To factor the numerator; $x^2 + 4x + 3$, find two numbers that have a product(multiply) of 3 and sum(add) to 4; **3 and 1 multiply to 3 and add to 4;**

$$\frac{(x+3)(x+1)}{(x+3)}$$

where $x + 3$ cancels in both the numerator and denominator. That leaves the final solution; $x + 1$

$x + 1$ Answer B Difficulty Level 5

2) When adding fractions, the first step is to convert every fraction into the same denominator. To convert fractions, find the least common denominator. The different denominators in this question are $3a$ and 2 , and the lowest factor they both divide into equally is $6a$. A good technique when adding fractions like these is to line the fractions up vertically.

$$\begin{array}{rcl} \frac{1}{3a} = \frac{?}{6a} ; 3a \text{ divides into } 6a, 2 \text{ times; } 2 \text{ times } 1 = 2; & & \frac{2}{6a} \\ + \frac{3a}{2} = \frac{?}{6a} ; 2 \text{ divides into } 6a, 3a \text{ times; } 3a \text{ times } 3a = 9a^2 & & \frac{9a^2}{6a} \\ \hline = \frac{3}{3a} = \frac{?}{6a} ; 3a \text{ divides into } 6a, 2 \text{ times; } 2 \text{ times } 3 = 6 & & = \frac{6}{6a} \end{array}$$

Finally, ignore the denominators, since they are all the same, and solve for a in the numerator; $2 + 9a^2 = 6$, $9a^2 = 4$, $a^2 = \frac{4}{9} = \pm \frac{2}{3}$. Since only positive $\frac{2}{3}$ is an answer choice; it is the correct answer choice.

Another way to answer this question correctly is to subtract $\frac{1}{3a}$

from both sides of the equation, then *cross multiply*.

$$\frac{1}{3a} + \frac{3a}{2} = \frac{3}{3a}$$

$$-\frac{1}{3a} \quad -\frac{1}{3a}$$

$$\frac{3a}{2} = \frac{2}{3a}$$

$$3a \times 3a = 9a^2; 2 \times 2 = 4, 9a^2 = 4, a^2 = \frac{4}{9}, a = \pm \frac{2}{3}$$

Taking the square root of both sides of an equation produces both a positive and a negative root or solution.

$\frac{2}{3}$ **Answer E** Difficulty Level 7

3) When solving for radicals, always isolate the radical. The radical is already isolated because it is by itself on the left side of the equation. To solve for x , start by squaring both sides of the equation; $(\sqrt{x-3})^2 = (5)^2$. When squaring an expression under a radical, the radical cancels with the square.
 $x - 3 = 25, x = 28$.

28 **Answer B** Difficulty Level 3

4) If there is a trinomial in the form $ax^2 + bx + c$ on the math ACT, factor, and it will almost always lead to the correct solution. To factor the trinomial $x^2 - 8x + 15$, find two numbers that have a product (multiply) of **15** and sum (add) to **-8**; -5 and -3 multiply to **15** and add to **-8**.

Set the factored form equal to zero; $(x - 5)(x - 3) = 0$.

Solve; $x - 5 = 0$ and $x - 3 = 0, x = 5$ and $x = 3$

3 and 5 **Answer C** Difficulty Level 6

5) To simplify the expression, notice that the numerator is a trinomial in the form $ax^2 + bx + c$ therefore, it may be factorable. To factor, always look for the greatest common factor. Since $6x^2 - 15x + 9$ has a GCF of 3, factor out the 3; $3(2x^2 - 5x + 3)$. There are numerous ways of factoring a trinomial with a **leading coefficient** not equal to one. **Use the guess and check method;**

$(2x \quad)(x \quad)$ since 2 is a prime number, the first terms must be **2x** and **x** to multiply to $2x^2$, this part must work if the trinomial is factorable. Next, find two numbers that

multiply to 3 and when the equation is foiled, produce the original equation. Guess 1: $(2x + 3)(x + 1)$ will not work because that would produce $2x^2 + 2x + 3x + 3 = 2x^2 + 5x + 3$ which has a + sign before 5x instead of a minus sign. The only other alternative $(2x - 3)(x - 1)$ does work. Foiling it will give the original equation after factoring out the 3. Finally, **cancel** the same terms in both the numerator and denominator; $\frac{3(2x-3)(\cancel{x-1})}{(\cancel{x-1})} = 3(2x - 3)$

3(2x - 3) Answer C Difficulty Level 7.5

6) When adding fractions, the first step is to convert every fraction into the same denominator. To convert the fractions, find the least common denominator of M , 10, and 5. LCD = $10M$. A good technique when adding fractions like these is to line the fractions up vertically.

$$\begin{aligned} \frac{4}{M} &= \frac{\quad}{10M} = \frac{40}{10M} \text{ (} M \text{ divides into } 10M, 10 \text{ times; } 10 \text{ times } 4 = 40\text{)} \\ + \frac{3}{10} &= \frac{\quad}{10M} = \frac{3M}{10M} \text{ (} 10 \text{ divides into } 10M, M \text{ times; } M \text{ times } 3 = 3M\text{)} \\ &= \frac{4}{5} = \frac{\quad}{10M} = \frac{8M}{10M} \text{ (} 5 \text{ divides into } 10M, 2M \text{ times; } 2M \text{ times } 4 = 8M\text{)} \end{aligned}$$

$\frac{40}{10M} + \frac{3M}{10M} = \frac{8M}{10M}$; since the denominators are all the same, solve the numerators to arrive at the correct solution;

$$40 + 3M = 8M, 5M = 40, M = 8.$$

8 Answer A Difficulty Level 7

7) To solve, move everything to one side of the equation and set it equal to zero. $4Q^2 + 3 = 12Q - 6$; subtract $12Q$ from both sides of the equation and add 6 to both sides; $4Q^2 - 12Q + 9 = 0$; Factor if possible, then solve for Q . There is not a common factor so, factor the trinomial; There are numerous different factoring methods. Since the **leading coefficient(4)** is composite and has two pairs of factors; **(4 and 1)(2 and 2)**, The guess and check method will produce either $4Q$ and Q as the leading coefficients of the two factors or $2Q$ and $2Q$. All the different possibilities with a negative sign are listed; Since the 9 is positive and the middle term of the trinomial is negative, the constants, when factoring, must both have a negative sign, meaning there is a minus sign in the middle of each term for each binomial.

$$(4Q - 3)(Q - 3) = 4Q^2 - 15Q + 9$$

$$(4Q - 9)(Q - 1) = 4Q^2 - 13Q + 9$$

$$(2Q - 3)(2Q - 3) = 4Q^2 - 12Q + 9$$

$$(2Q - 9)(2Q - 1) = 4Q^2 - 20Q + 9$$

Solve for Q ; $2Q - 3 = 0$, $2Q = 3$, $Q = \frac{3}{2}$

$\frac{3}{2}$ Answer D Difficulty Level 7

8) When adding fractions, it is usually easiest to line them up vertically. Next, find a common denominator, preferably the least common denominator. The LCD is the lowest common denominator that every other denominator divides into equally. In this question, the two denominators are $3A$ and $4A$, which have a least common denominator of $12A$. Convert both fractions with the new denominator, then add the numerators;

$$\frac{4}{3A} = \frac{\quad}{12A} = \frac{16}{12A} \text{ (3A divides into 12A 4 times, 4 times 4 = 16)}$$

$$+ \frac{3}{4A} = \frac{\quad}{12A} = \frac{9}{12A} \text{ (4A divides into 12A 3 times, 3 times 3 = 9)}$$

$$= \frac{25}{12A}$$

$\frac{25}{12A}$ Answer B Difficulty Level 6